Open Critical Mapping Student Workshop

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Packages

Let's start by installing/loading packages (as needed) and libraries:

```
# Load necessary libraries
library(tidycensus)
library(tidyverse)
library(sf)
library(viridis)
library(mapview)
```

U.S. Census API key

To access the census data for mapping, please sign up for an API key using this link. You should receive an API key in your email. Place it in the code chunk as indicated below.

Introducing Your Variable

My variable are renters that occupy households in Fulton County, GA.

(B25003_003E / B25003_001E)*100 # Mapping

```
acs_vars <- load_variables(2023, dataset = "acs5")

ga_OHU <- get_acs(
  geography = "county",
  variables = "B25003_001E",
  state = "GA",
  year = 2023
)
```

```
## Getting data from the 2019-2023 5-year ACS
```

```
ga_OHU
```

```
## # A tibble: 159 x 5
##    GEOID NAME                      variable   estimate   moe
##    <chr> <chr>                     <chr>         <dbl> <dbl>
##  1 13001 Appling County, Georgia   B25003_001     7037   291
##  2 13003 Atkinson County, Georgia  B25003_001     3189   109
##  3 13005 Bacon County, Georgia     B25003_001     4013   227
##  4 13007 Baker County, Georgia     B25003_001     1114   125
##  5 13009 Baldwin County, Georgia   B25003_001    16075   517
##  6 13011 Banks County, Georgia     B25003_001     6787   143
##  7 13013 Barrow County, Georgia    B25003_001    29770   293
##  8 13015 Bartow County, Georgia    B25003_001    39664   688
##  9 13017 Ben Hill County, Georgia  B25003_001     7162   246
## 10 13019 Berrien County, Georgia   B25003_001     6881   223
## # i 149 more rows
```

```
fulton_OHU <- get_acs(
  geography = "tract",
  variables = "B25003_001E",
  state = "GA",
  county = "Fulton",
  year = 2023,
  geometry = TRUE
)
```

```
## Getting data from the 2019-2023 5-year ACS

## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions

##   |                                                                              |
```

```
fulton_OHU
```

```
## Simple feature collection with 327 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -84.85071 ymin: 33.50251 xmax: -84.09769 ymax: 34.18629
## Geodetic CRS:  NAD83
## First 10 features:
##          GEOID                                          NAME   variable estimate
## 1  13121010601 Census Tract 106.01; Fulton County; Georgia B25003_001     1471
## 2  13121008301  Census Tract 83.01; Fulton County; Georgia B25003_001     1135
## 3  13121002400     Census Tract 24; Fulton County; Georgia B25003_001     1054
## 4  13121006400     Census Tract 64; Fulton County; Georgia B25003_001      443
## 5  13121002900     Census Tract 29; Fulton County; Georgia B25003_001     1420
## 6  13121980000   Census Tract 9800; Fulton County; Georgia B25003_001        0
## 7  13121003700     Census Tract 37; Fulton County; Georgia B25003_001      233
## 8  13121000700      Census Tract 7; Fulton County; Georgia B25003_001      392
## 9  13121008203  Census Tract 82.03; Fulton County; Georgia B25003_001     1228
## 10 13121009604  Census Tract 96.04; Fulton County; Georgia B25003_001     2078
##    moe                       geometry
## 1  300 MULTIPOLYGON (((-84.4686 33...
## 2  246 MULTIPOLYGON (((-84.47144 3...
```

```
## 3  174 MULTIPOLYGON (((-84.43354 3...
## 4   97 MULTIPOLYGON (((-84.37848 3...
## 5  135 MULTIPOLYGON (((-84.37433 3...
## 6   14 MULTIPOLYGON (((-84.4403 33...
## 7  102 MULTIPOLYGON (((-84.41132 3...
## 8   50 MULTIPOLYGON (((-84.43237 3...
## 9  306 MULTIPOLYGON (((-84.4925 33...
## 10 214 MULTIPOLYGON (((-84.38733 3...
```

```
ga_OHU_sf <- get_acs(
  geography = "county",
  variables = "B25003_001E",
  state = "GA",
  year = 2023,
  geometry = TRUE
)
```

```
## Getting data from the 2019-2023 5-year ACS

## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions

##   |                                                                              |
```

```
ga_OHU_sf
```

```
## Simple feature collection with 159 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -85.60516 ymin: 30.35785 xmax: -80.84038 ymax: 35.00124
## Geodetic CRS:  NAD83
## First 10 features:
##    GEOID                       NAME   variable estimate  moe
## 1  13185   Lowndes County, Georgia B25003_001    44928  565
## 2  13127     Glynn County, Georgia B25003_001    34625  774
## 3  13051   Chatham County, Georgia B25003_001   119241 1147
## 4  13179   Liberty County, Georgia B25003_001    23406  485
## 5  13073  Columbia County, Georgia B25003_001    52083  969
## 6  13137 Habersham County, Georgia B25003_001    16048  345
## 7  13089    DeKalb County, Georgia B25003_001   284681 1814
## 8  13157   Jackson County, Georgia B25003_001    26976  335
## 9  13267  Tattnall County, Georgia B25003_001     7551  302
## 10 13031   Bulloch County, Georgia B25003_001    30276  612
##                          geometry
## 1  MULTIPOLYGON (((-83.48448 3...
## 2  MULTIPOLYGON (((-81.78263 3...
## 3  MULTIPOLYGON (((-81.3917 32...
## 4  MULTIPOLYGON (((-81.8244 32...
## 5  MULTIPOLYGON (((-82.44285 3...
## 6  MULTIPOLYGON (((-83.6816 34...
## 7  MULTIPOLYGON (((-84.34984 3...
## 8  MULTIPOLYGON (((-83.81768 3...
## 9  MULTIPOLYGON (((-82.23371 3...
## 10 MULTIPOLYGON (((-82.02684 3...
```

```r
fulton_OHU_sf <- get_acs(
  geography = "tract",
  variables = "B25003_001E",
  state = "GA",
  county = "Fulton",
  year = 2023,
  geometry = TRUE
)
```

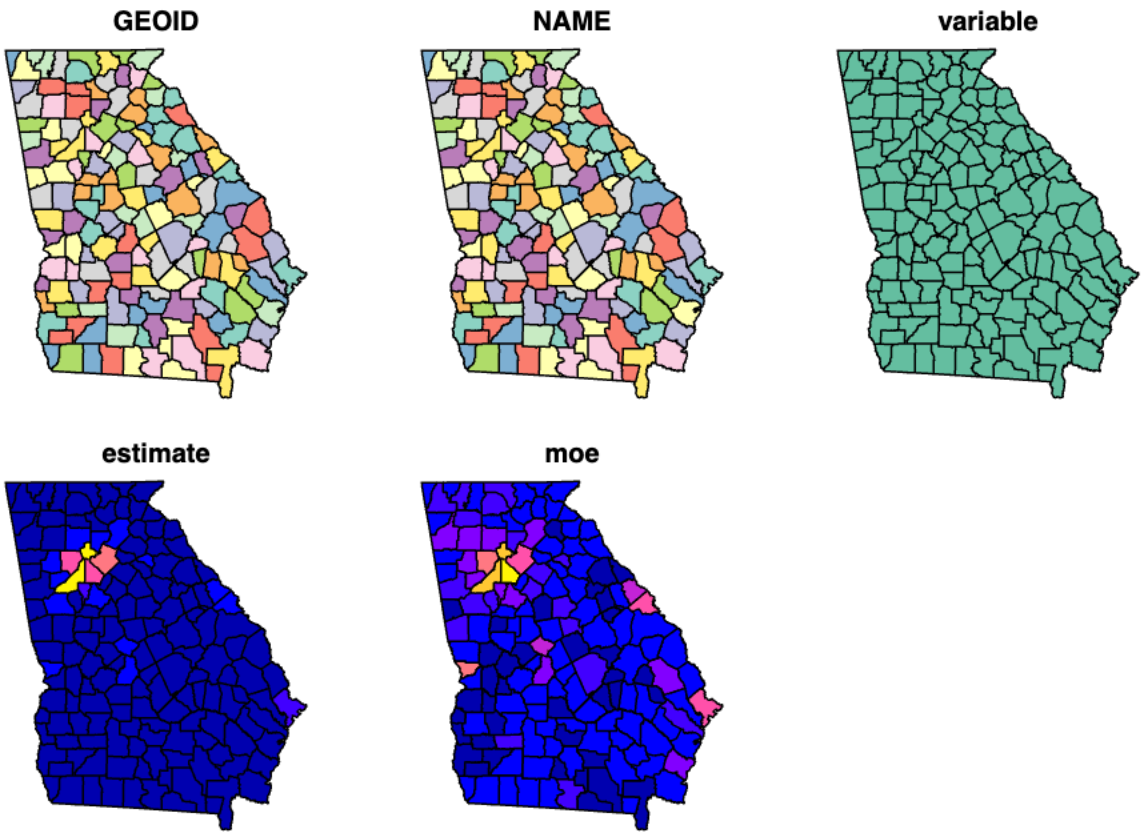
```
## Getting data from the 2019-2023 5-year ACS

## Downloading feature geometry from the Census website.  To cache shapefiles for use in future session
```

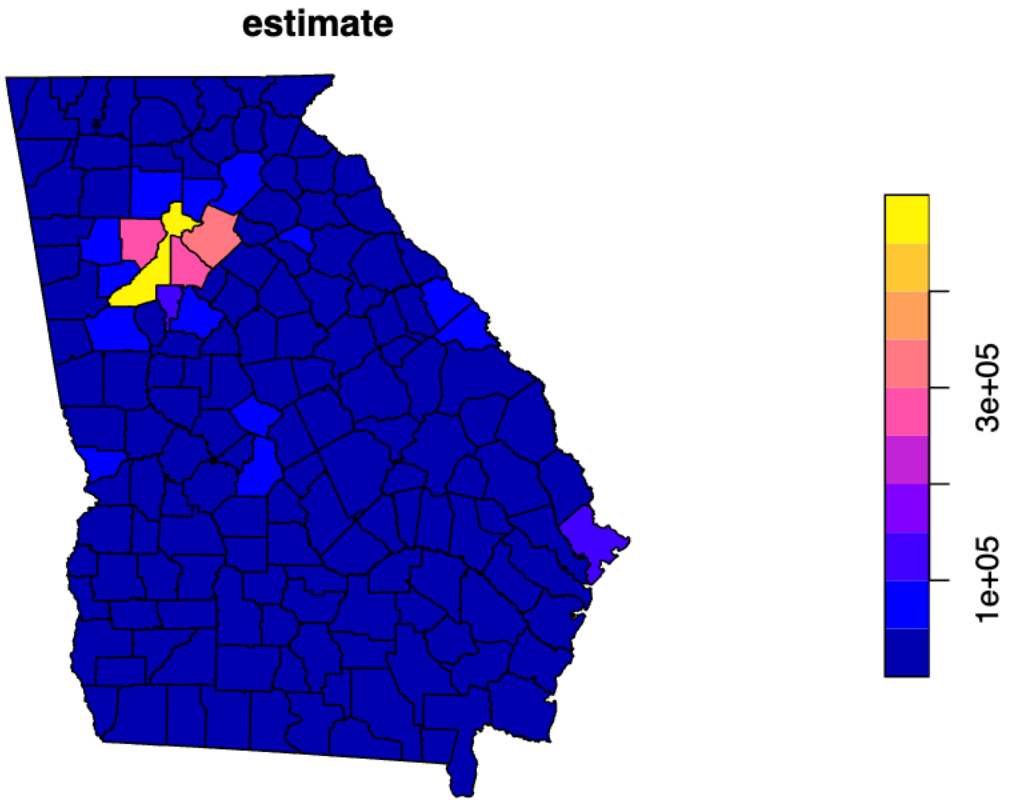
```r
fulton_OHU_sf
```

```
## Simple feature collection with 327 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -84.85071 ymin: 33.50251 xmax: -84.09769 ymax: 34.18629
## Geodetic CRS:  NAD83
## First 10 features:
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## 9  306 MULTIPOLYGON (((-84.4925 33...
## 10 214 MULTIPOLYGON (((-84.38733 3...
```

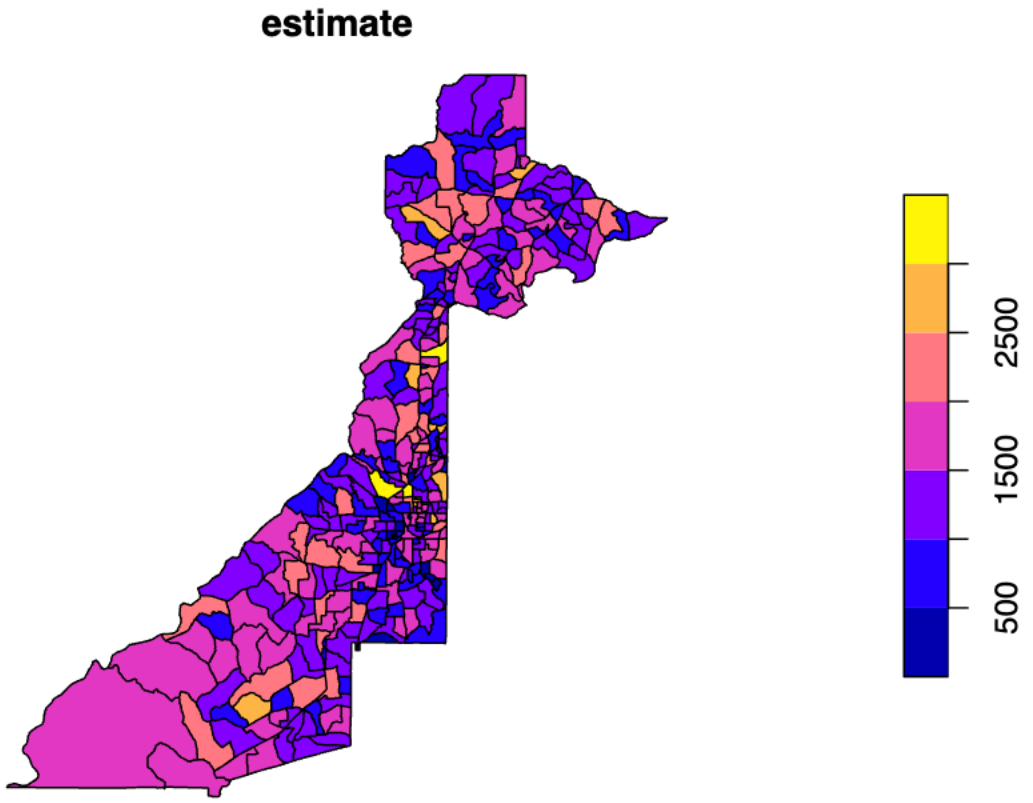
```r
plot(ga_OHU_sf)
```



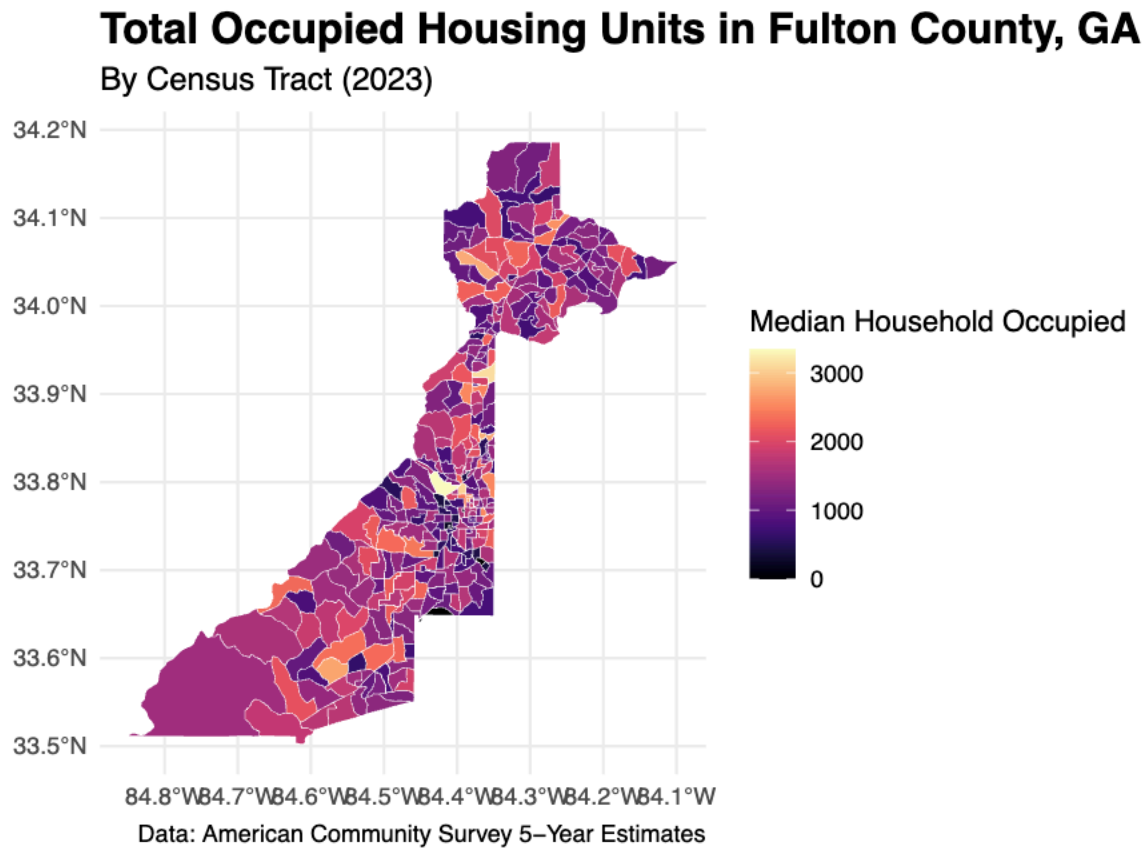

```
plot(ga_OHU_sf['estimate'])
```




```
plot(fulton_OHU_sf['estimate'])
```




```
ggplot(fulton_OHU_sf) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.1) +
  scale_fill_viridis(option = "magma", name = "Median Household Occupied") +
  labs(
    title = "Total Occupied Housing Units in Fulton County, GA",
    subtitle = "By Census Tract (2023)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```




```
mapview(
  fulton_OHU_sf,
  zcol = "estimate"
)
```

```
fulton_renters <- get_acs(
  geography = "tract",
  variables = "B25003_003E", # Renters Occupied
  state = "GA",
  county = "Fulton",
  year = 2023,
  geometry = T
)
```

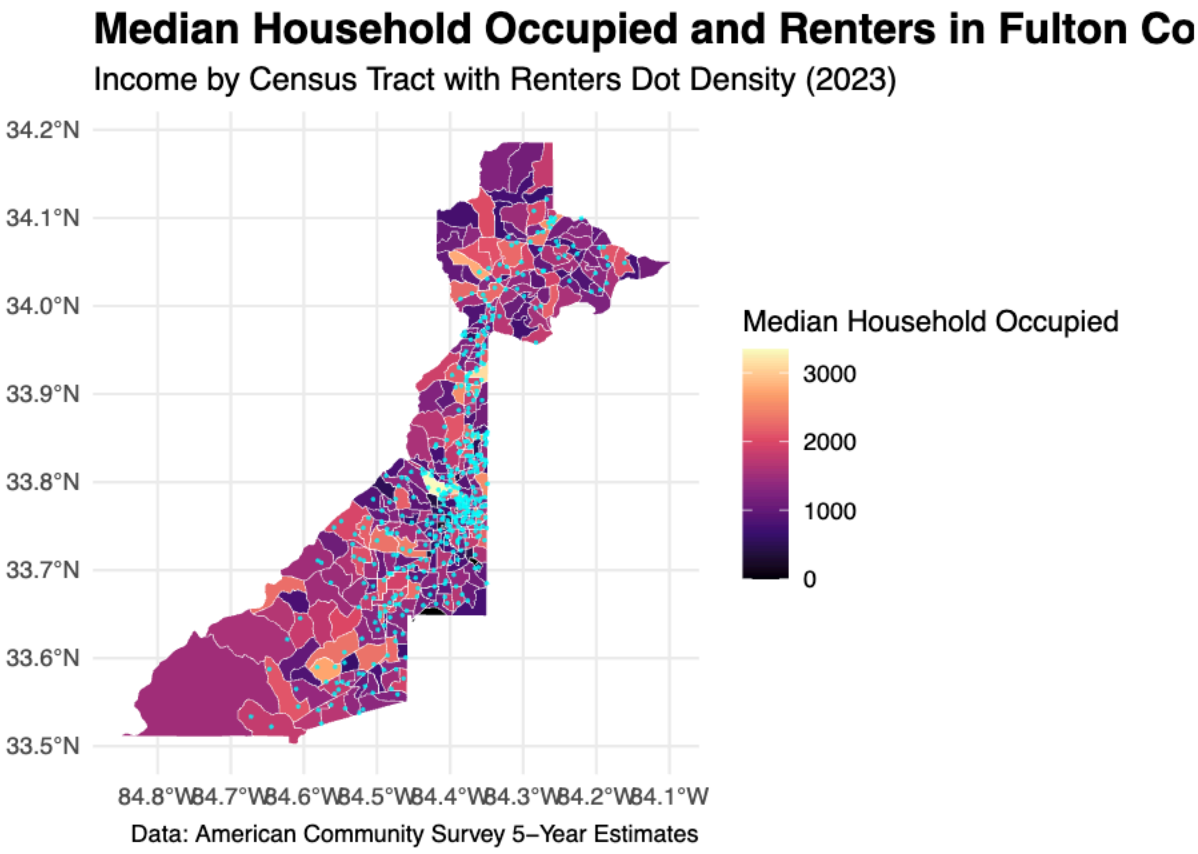
```
## Getting data from the 2019-2023 5-year ACS

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```

```
renters_dots <- as_dot_density(
  fulton_renters,
  value = "estimate",
  values_per_dot = 500
)
```

```
mapview(
  fulton_OHU_sf, # median household occupied
  zcol = "estimate",
  alpha.regions = 0.75
)+
  mapview(renters_dots, cex = 0.01,
          layer.name = " Renters <br>1 dot = 500 people", # Renters
          col.regions = "NA", color = "red")
```

```
ggplot() +
  geom_sf(
    data = fulton_OHU_sf,
    aes(fill = estimate),
    color = "white",
    linewidth = 0.1
  ) +
  geom_sf(
    data = renters_dots,
    color = "cyan",
    size = 0.25,
    alpha = 0.7,
    inherit.aes = FALSE
  ) +
  scale_fill_viridis_c(
    option = "magma",
    name = "Median Household Occupied"
  ) +
  labs(
    title = "Median Household Occupied and Renters in Fulton County, GA",
    subtitle = "Income by Census Tract with Renters Dot Density (2023)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```


Median Household Occupied and Renters in Fulton Co
Income by Census Tract with Renters Dot Density (2023)
34.2°N
34.1°N
34.0°N
33.9°N
33.8°N
33.7°N
33.6°N
33.5°N
84.8°W
84.7°W
84.6°W
84.5°W
84.4°W
84.3°W
84.2°W
84.1°W
Median Household Occupied
3000
2000
1000
0
Data: American Community Survey 5–Year Estimates